16 Berenika wants to buy 35 T-shirts.

Each T-shirt costs £5.80
Berenika does the calculation $40 \times 6 = 240$ to estimate the cost of 35 T-shirts.

(a) Explain how Berenika's calculation shows the actual cost will be less than £240

Both factors were rounded up: 35 to 40 and £5.80 to £6.

Their product therefore overestimates the actual cost.

1 mark
Explain that both positive factors were increased

(1)

There is a special offer.

T-shirts £5.80 each. Buy 30 or more T-shirts. Get 10% off the total cost.

(b) Work out the actual cost of buying 35 T-shirts using the special offer.

First calculate 58 × 35.

	50	+8
30	1500	+240
+5	+250	+40

N = number; GBP = pounds.

N	GBP
1	5.8
35	203

(×35 from 1 to 35; ×35 from 5.8 to 203)

1500 + 240 + 250 + 40 = 2030

£5.80 × 35 = £203

1 mark
Find the cost before the discount

P = percentage.

P	GBP
100	203
10	20.3
90	182.7

(÷10 from 100 to 10; ×9 from 10 to 90; ÷10 from 203 to 20.3; ×9 from 20.3 to 182.7)

1 mark
Find 10% of £203

1 mark
Pay 90% after the 10% discount

1 mark
Correct cost to the nearest penny

182.70

£.....

(4)

(Total for Question 16 is 5 marks)

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